Digital signal processing

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1 Conventions

Let $f : \mathbb{R} \to \mathbb{C}$ be a 2π-periodic, integrable function. The Fourier coefficients of f are defined by

$$\hat{f}(k) = \frac{1}{2\pi} \int_0^{2\pi} f(x)\, e^{-ikx}\, \mathrm{d}x, \quad k \in \mathbb{Z}.$$

The Fourier series of f is given by

$$f(x) \;=\; \sum_{k \in \mathbb{Z}} \hat{f}(k)\, e^{ikx}$$

For 2π-periodic functions $f, g \colon \mathbb{R} \to \mathbb{C}$ we define the circular convolution by

$$(f * g)(x) \;=\; \int_0^{2\pi} f(y)\, g(x-y)\, \mathrm{d}y \tag{1}$$

The root of unity:

$$\omega_N = e^{\frac{2\pi i}{N}}$$

2 DFT

Lemma 2.1. *The following holds for roots of unity:*

2.1.1 $\omega_N^{kN} = 1 \; \forall k \in \mathbb{Z}$

2.1.2 N *even.* $\omega_{N/2} = \omega_N^2$

2.1.3 $\overline{\omega_N} = \omega_N^{-1}$

2.1.4 $\omega_N^{kN/2} = (-1)^k \; \forall k \in \mathbb{Z}$

Proof.

1. $$\omega_N^N = \left(e^{\frac{2\pi i}{N}}\right)^N = e^{\frac{2\pi i N}{N}} = e^{2\pi i} = \cos(2\pi) + i\sin(2\pi) = 1$$
$$\omega_N^{kN} = \left(\omega_N^N\right)^k = 1^k = 1$$

2. $\omega_{N/2} = e^{\frac{2\pi i}{N/2}} = e^{2\cdot\frac{2\pi i}{N}} = \left(e^{\frac{2\pi i}{N}}\right)^2 = \omega_N^2$

3. $\overline{\omega_N} = \overline{e^{\frac{2\pi i}{N}}} = e^{-\frac{2\pi i}{N}} = \left(e^{\frac{2\pi i}{N}}\right)^{-1} = \omega_N^{-1}$

4. $\omega_N^{k\cdot N/2} = \left(e^{2\pi i\frac{N/2}{N}}\right)^k = \left(e^{\pi i}\right)^k = (-1)^k$

□

Suppose we have $f : \mathbb{R} \to \mathbb{C}$ which is 2π-periodic. Consider the values of this function at points $\frac{0\cdot 2\pi}{N}$, $\frac{1\cdot 2\pi}{N}$, ..., $\frac{(N-1)\cdot 2\pi}{N}$:

$$x_j := \frac{j\cdot 2\pi}{N}$$

$$y_j := f(x_j) = \sum_{k=-\infty}^{\infty} \hat{f}(k)e^{ikx_j} = \sum_{k=-\infty}^{\infty} \hat{f}(k)e^{2\pi i\frac{kj}{N}} = \sum_{k=-\infty}^{\infty} \hat{f}(k)\omega_N^{kj}$$

By Lemma 2.1.1 we have that $\omega_N^{(k+nN)j} = \omega_N^{kj}\omega_N^{(nj)N} = \omega_N^{kj}$
By grouping the terms with $k_1 = k_2 \mod N$ and obtain the following:

$$y_j = \sum_{k=0}^{N-1} \omega_N^{kj} \left(\sum_{n=-\infty}^{\infty} \hat{f}(k+nN)\right) = \sum_{k=0}^{N-1} \omega_N^{kj} z_k \tag{2}$$

We define the matrix $F_{kj} = \frac{1}{\sqrt{N}}\omega_N^{kj}$. Then:

$$y_j = \sqrt{N}(Fz)_j \Rightarrow y = \sqrt{N}Fz \Rightarrow z = \frac{1}{\sqrt{N}}F^{-1}y = Ty \tag{3}$$

Later we use the vector z to indicate the output of the DFT, namely $F^{-1}y$

Theorem 2.1. *The matrix $F_{kj} = \frac{1}{\sqrt{N}}\omega_N^{kj}$ is unitary*

Proof.

$$(F\overline{F^T})_{jl} = \frac{1}{\sqrt{N}}\overline{\left(\frac{1}{\sqrt{N}}\right)}\sum_{k=0}^{N-1} F_{jk}(\overline{F^T})_{kl} = \frac{1}{N}\sum_{k=0}^{N-1} F_{jk}\overline{F_{lk}} =$$
$$\frac{1}{N}\sum_{k=0}^{N-1}\omega_N^{jk}\overline{\omega_N^{lk}} \underset{(a)}{=} \frac{1}{N}\sum_{k=0}^{N-1}\omega_N^{jk}\omega_N^{-lk} = \frac{1}{N}\sum_{k=0}^{N-1}\omega_N^{(j-l)k}$$

In (a) we used the identity $\overline{\omega_N^{lk}} = \omega_N^{-lk}$ from Lemma 2.1.3

1. $j \neq l$. Using the formula for sum of geometric series:

$$\sum_{k=0}^{N-1}\omega_N^{(j-l)k} = \frac{1-\left(\omega_N^{j-l}\right)^N}{1-\omega^{j-l}} = \frac{1-\left(\omega_N^N\right)^{j-l}}{1-\omega^{j-l}} \underset{(a)}{=} \frac{1-1^{j-l}}{1-\omega^{j-l}} = 0$$

   In (a) we used the identity $\omega_N^N = 1$ from Lemma 2.1.1. The condition $j \neq l$ ensures that the denominator is non-zero, so the expression is well-defined.

2. $j = l$. Then:

$$\sum_{k=0}^{N-1}\omega_N^{(j-l)k} = \sum_{k=0}^{N-1} 1 = N \Rightarrow (F\overline{F^T})_{jl} = \frac{1}{N}N = 1$$

The matrix $FF^* = I$, so F unitary and $F^{-1} = \overline{F^T}$
The entries of F^{-1} are given by $F_{ij}^{-1} = (\overline{F_{ji}}) = \frac{1}{\sqrt{N}}\overline{\left(\omega_N^{ji}\right)} = \frac{1}{\sqrt{N}}\omega_N^{-ij}$. $\square$

Using the result of Theorem 2.1 and equations (3), (2) we define the discrete Fourier transform:

$$z_j = (F^{-1}y)_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kj} y_k \tag{DFT}$$

$$y_j = (Fz)_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{kj} z_k \tag{IDFT}$$

Remark 2.1. Now we make two remarks about the DFT structure:

2.1.1 Firstly using DFT of size n we are unable to detect the frequencies above the (n-1)-th one. In fact as the equation (2) suggests, the frequency of the form $k + nN$ with $k \in 0, ..., N-1$ contributes to the entry with index k. However the maximum frequency human ear can process is approximately 20000, so this issue is handled by choosing an appropriate sample rate, that is usually 44100hz, which is double the maximum frequency which can be detected. The exact reason for it such choice is discussed later. The following plot illustrates the issue discussed (so-called aliasing):

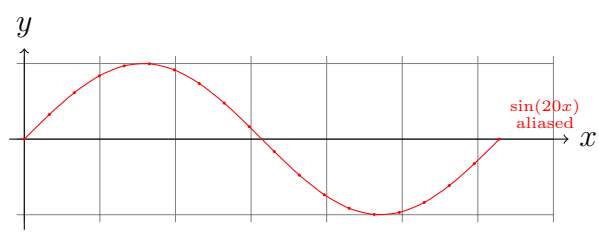

Figure 1: 20 discrete points

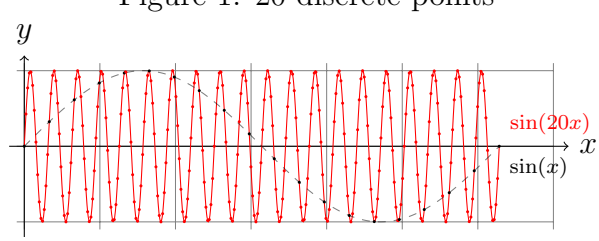

Figure 2: 286 discrete points

2.1.2 The meaning of the phase is often overlooked in the literature, so we consider an example: Let $y = e_k + e_{N-k}$. Then:

$$Ty = T_{*k} + T_{*(N-k)} = \begin{pmatrix} \omega_N^0 \\ \omega_N^{-k} \\ \vdots \\ \omega_N^{-k(N-1)} \end{pmatrix} + \begin{pmatrix} \omega_N^0 \\ \omega_N^{-(N-k)} \\ \vdots \\ \omega_N^{-(N-k)(N-1)} \end{pmatrix}$$

$$\underset{\text{Lemma 2.1.1}}{=} \begin{pmatrix} \omega_N^0 + \omega_N^0 \\ \omega_N^{-k} + \omega_N^k \\ \vdots \\ \omega_N^{-k(N-1)} + \omega_N^{k(N-1)} \end{pmatrix} = \begin{pmatrix} 2\cos(0) \\ 2\cos\left(\frac{2\pi k}{N}\right) \\ \vdots \\ 2\cos\left(\frac{2\pi k(N-1)}{N}\right) \end{pmatrix}$$

As we see from the computation, the output is the function $\cos(kx)$ discretized at points $x_j = \frac{2\pi j}{N}$ for $j \in \{0, \ldots, N-1\}$. What happens if we multiply the positive frequency component of y by ω_N^{-k} and the negative one by ω_N^k?

$$y^* = \omega_N^{-k} e_k + (\omega_N^k e_{N-k})$$

$$Ty^* = \begin{pmatrix} \omega_N^{-k}\omega_N^0 + \omega_N^k\omega_N^0 \\ \omega_N^{-k}\omega_N^{-k} + \omega_N^k\omega_N^k \\ \vdots \\ \omega_N^{-k}\omega_N^{-k(N-1)} + \omega_N^k\omega_N^{k(N-1)} \end{pmatrix}$$

$$= \begin{pmatrix} \omega_N^{-k} + \omega_N^k \\ \omega_N^{-2k} + \omega_N^{2k} \\ \vdots \\ \omega_N^{-kN} + \omega_N^{kN} \end{pmatrix} = \begin{pmatrix} 2\cos\left(\frac{2\pi k}{N}\right) \\ 2\cos\left(\frac{2\pi\cdot 2k}{N}\right) \\ 2\cos\left(\frac{2\pi\cdot 3k}{N}\right) \\ \vdots \\ 2\cos\left(\frac{2\pi kN}{N}\right) \end{pmatrix}$$

We effectively moved the phase of the cosine by $\frac{2\pi}{N}$, since

$$(Ty^*)_j = 2\cos\left(\frac{2\pi k(j+1)}{N}\right) = 2\cos\left(k\left(\frac{2\pi j}{N} + \frac{2\pi}{N}\right)\right)$$
$$= 2\cos\left(k\left(x_j + \frac{2\pi}{N}\right)\right)$$

not hard yet difficult to see by yourself (at least for me)

Theorem 2.2 (Real signal conjugate symmetry)**.** *Let N be even. The following are equivalent:*

1. $y = \overline{y}, y \in \mathbb{C}^N$
2. $z_j = \overline{z_{N-j}} \quad \forall j \in \{1, \dots, N/2 - 1\}$ *and* $z_0, z_{N/2} \in \mathbb{R}$

Proof.
"(1) $\Rightarrow$ (2)" Using the formula for DFT

$$z_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kj} y_k = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \overline{\omega_N^{kj}} y_k \underset{y=\overline{y}}{=} \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \overline{\omega_N^{kj} y_k} =$$

$$\overline{\left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{kj} y_k \right)} \underset{\omega_N^{kN}=1}{=} \overline{\left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kN} \omega_N^{kj} y_k \right)} =$$

$$\overline{\left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-k(N-j)} y_k \right)} = \overline{z_{N-j}}$$

$$z_0 = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-k \cdot 0} y_k = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} y_k \in \mathbb{R}$$

By Lemma 2.1.4:

$$z_{N/2} = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kN/2} y_k = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} (-1)^k y_k \in \mathbb{R}$$

"(2) $\Rightarrow$ (1)" Using the formula for IDFT

$$y_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{kj} z_k = \frac{1}{\sqrt{N}} \left(z_0 + (-1)^j z_{N/2} + \sum_{k=1}^{N/2-1} \omega_N^{kj} z_k + \sum_{l=N/2+1}^{N-1} \omega_N^{lj} z_l \right) \tag{4}$$

$$\sum_{k=1}^{N/2-1} \omega_N^{kj} z_k + \sum_{l=N/2+1}^{N-1} \omega_N^{lj} z_l \underset{l'=N-l}{=} \sum_{k=1}^{N/2-1} \omega_N^{kj} z_k + \sum_{l=1}^{N/2-1} \omega_N^{(N-l)j} z_{N-l} =$$

$$\sum_{k=1}^{N/2-1} \left(\omega_N^{kj} z_k + \omega_N^{(N-k)j} z_{N-k} \right) = \sum_{k=1}^{N/2-1} \left(\omega_N^{kj} z_k + \overline{\omega_N^{kj} z_k} \right) \in \mathbb{R}$$

By plugging this expression back to (4) we get:

$$(4) = \frac{1}{\sqrt{N}} \left(z_0 + (-1)^j z_{N/2} + \sum_{k=1}^{N/2-1} \left(\omega_N^{kj} z_k + \overline{\omega_N^{kj} z_k} \right) \right) \in \mathbb{R}$$

since all the summands and the factor in front are real. $\square$

3 FFT

Theorem 3.1. *Let $y \in \mathbb{C}^N$ and N is even. $K = N/2$. Let:*

$$y^{(1)} = (y_0, y_2, ..., y_{2K-2})^T \quad y^{(2)} = (y_1, y_3, ..., y_{2K-1})^T$$
$$z^{(1)} = F_K^{-1} y^{(1)} \quad z^{(2)} = F_K^{-1} y^{(2)}$$

Then:

$$z_j = \frac{1}{\sqrt{2}} \left(z_j^{(1)} + \omega_N^{-j} z_j^{(2)} \right) \ \forall j \in \{0, \dots, K-1\}$$
$$z_{j+K} = \frac{1}{\sqrt{2}} \left(z_j^{(1)} - \omega_N^{-j} z_j^{(2)} \right) \ \forall j \in \{0, \dots, K-1\}$$

where $z = F_N^{-1} y$

Proof.

$$\begin{aligned} z_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kj} y_k = \frac{1}{\sqrt{N}} \left(\sum_{\substack{k=0 \\ k \text{ even}}}^{N-1} \omega_N^{-kj} y_k + \sum_{\substack{k=0 \\ k \text{ odd}}}^{N-1} \omega_N^{-kj} y_k \right) = \\ \frac{1}{\sqrt{N}} \left(\sum_{k=0}^{K-1} \omega_N^{-2kj} y_{2k} + \sum_{k=0}^{K-1} \omega_N^{-(2k+1)j} y_{2k+1} \right) \underset{\text{Lemma 2.1.2}}{=} \\ \frac{1}{\sqrt{N}} \left(\sum_{k=0}^{K-1} \omega_K^{-kj} y_{2k} + \omega_N^{-j} \sum_{k=0}^{K-1} \omega_K^{-kj} y_{2k+1} \right) = \\ \frac{1}{\sqrt{2}\sqrt{K}} \left(\sum_{k=0}^{K-1} \omega_K^{-kj} y_{2k} + \omega_N^{-j} \sum_{k=0}^{K-1} \omega_K^{-kj} y_{2k+1} \right) \end{aligned} \tag{5}$$

For $j \in \{0, \dots, K-1\}$ we get the claim immediately. We now focus on the case, when $j = j' + K \in \{K, \dots, N-1\}$. The equation (5) becomes:

$$\begin{aligned} \sqrt{2K} z_j = \sum_{k=0}^{K-1} \omega_K^{-k(j'+K)} y_{2k} + \omega_N^{-(j'+K)} \sum_{k=0}^{K-1} \omega_K^{-k(j'+K)} y_{2k+1} = \\ \sum_{k=0}^{K-1} \omega_K^{-kj'} y_{2k} + \omega_N^{-j'} \omega_N^{-K} \sum_{k=0}^{K-1} \omega_K^{-kj'} y_{2k+1} \underset{\text{Lemma 2.1.4}}{=} \\ \sum_{k=0}^{K-1} \omega_K^{-kj'} y_{2k} - \omega_N^{-j'} \sum_{k=0}^{K-1} \omega_K^{-kj'} y_{2k+1} \end{aligned}$$

which proves the second part of the claim. $\square$

By extending the idea of the idea of the Theorem 3.1 we formulate the following algorithm:

Algorithm 3.1.
Input: $y \in \mathbb{C}^N$, N is a power of two, $n > 1$
Procedure:
if $N > n$:
Get $z^{(1)}$, $z^{(2)}$ by calling Algorithm 3.1 on $y^{(1)}$, $y^{(2)}$. This is well defined since $N/2$ is a power of two

$$z_j = \frac{1}{\sqrt{2}} \left(z_j^{(1)} + \omega_N^{-j} z_j^{(2)} \right) \ \forall j \in \{0, \ldots, K-1\}$$

$$z_{j+K} = \frac{1}{\sqrt{2}} \left(z_j^{(1)} - \omega_N^{-j} z_j^{(2)} \right) \ \forall j \in \{0, \ldots, K-1\}$$

return z
else:
return $F^{-1}y$ (see the expression DFT)

4 Finite impulse response filters

4.1 Convolution theorem

Lemma 4.1. *Let $f : \mathbb{R} \to \mathbb{R}$ have countably many discontinuities. Then f is measurable*

Proof. 1. Let $D \subset \mathbb{R}$ be the set of all discontinuities of f. We have that D has countably many elements. For any set $M \in \mathcal{B}$ we have the following:

$$\begin{aligned} f^{-1}(M) &= \left(f^{-1}(M) \cap D\right) \cup \left(f^{-1}(M) \setminus D\right) \\ &= \left(f^{-1}(M) \cap D\right) \cup \left(\left(f|_{\mathbb{R}\setminus D}\right)^{-1}(M)\right) \end{aligned} \tag{6}$$

We have $(f^{-1}(M) \cap D) \subset D$, so it is countable and hence is Borel measurable, $f|_{\mathbb{R}\setminus D}$ is a measurable mapping since $\mathbb{R} \setminus D$ is a Borel set and the restriction is continuous on its domain. Thus, $f^{-1}(M)$ is the union of two measurable sets, so is measurable as well. That proves measurability of f.

□

Remark 4.1. The statement of the Lemma 4.1 can be relaxed to the set of discontinuities having measure zero. Let F be a generalized distribution function (monotonous and right continuous). Then the following mapping defines an outer measure:

$$\mu^* : \mathcal{P}(\mathbb{R}) \to [0; \infty], A \mapsto \\ \inf\left\{\sum_{n=1}^{\infty} F(b_n) - F(a_n) : A \subset \bigcup_{n=1}^{\infty}(a_n; b_n],\ a_n < b_n\ \forall n \in \mathbb{N}\right\}$$

The Lebesgue measure is $\mu = \mu^*\big|_{\mathcal{B}}$ for $F(x) = x$. However we know that $\mathcal{B} \subset \mathcal{L} := \{A \subset \mathbb{R} : A \text{ is } \mu^* \text{ meaurable}\}$. However we cannot guarantee $\mathcal{B} = \mathcal{L}$ ($\mathcal{L}$ is a sigma algebra by Caratheodori restriction theorem)

Consider a set $N \subset \mathbb{R}$ with $\mu^*(N) = 0$. For any $B \subset \mathbb{R}$:

$$\mu^*(B) = \mu^*((B \cap N) \cup (B \cap N^C)) \leq \mu^*(B \cap N) + \mu^*(B \cap N^C)$$

by sigma subadditivity of μ^*. Moreover by monotonicty of μ^* we have that $\mu^*(B \cap N) \leq \mu^*(N) = 0$, so $\mu^*(B \cap N^C) \geq \mu^*(B)$. By applying monotonicty

again $\mu^*(B \cap N^C) \leq \mu^*(B)$, so $\mu^*(B \cap N^C) = \mu^*(B)$. By combining those results together we obtain $\mu^*(B) = \mu^*(B \cap N) + \mu^*(B \cap N^C)$, so $N \in \mathcal{M}$

Now take a look at (6):

$$f^{-1}(M) = \left(f^{-1}(M) \cap D\right) \cup \left(\left(f|_{\mathbb{R}\setminus D}\right)^{-1}(M)\right)$$

The set $f^{-1}(M) \cap D \subset D$, so it is a null set and hence is in $\mathcal{L}$. Also we know that $\left(f|_{\mathbb{R}\setminus D}\right)^{-1}(M) \subset \mathcal{B} \subset \mathcal{L}$, which implies $f^{-1}(M) \subset \mathcal{L}$. Therefore f is $\mathcal{L} \to \mathcal{B}$ measurable.

Theorem 4.1 (Convolution theorem)**.** *Let $f, g : \mathbb{R} \to \mathbb{R}$ be 2π periodic piecewise continuous and bounded. Then:*

$$\frac{1}{2\pi}\widehat{f * g}(k) = \hat{f}(k)\hat{g}(k)$$

Proof.

$$\begin{aligned} 2\pi \cdot \widehat{f * g}(k) &= \int_0^{2\pi} f * g(x) e^{-ikx} \,\mathrm{d}x \\ &\underset{(1)}{=} \int_0^{2\pi} \left(\int_0^{2\pi} f(y) g(x-y) \,\mathrm{d}y\right) \cdot e^{-ikx} \,\mathrm{d}x \\ &= \int_0^{2\pi} \int_0^{2\pi} f(y) g(x-y) e^{-ikx} \,\mathrm{d}y \,\mathrm{d}x \end{aligned} \tag{7}$$

Let $F(x, y) = f(y)g(x-y)e^{-ikx}$. We have that both f, g are measurabe by Lemma 4.1 since they have finitely many discontinuities. Therefore $\mathrm{Re}(F) = f(\pi_y(x, y))g(x - y)\cos(kx)$ and $\mathrm{Im}(F) = -f(\pi_y(x, y))g(x - y)\sin(kx)$ are $\mathcal{B}_2 \to \mathcal{B}$ measurable.

$$|\mathrm{Re}(F(x, y))|, |\mathrm{Im}(F(x, y))| \leq |F(x, y)| = |f(y)g(x - y)| \leq C < \infty$$

since both f and g are bounded.

$$\begin{aligned} \int_{[0,2\pi]^2} |\mathrm{Re}(F)(x, y)| \,\mathrm{d}\lambda^2(x, y) &\leq \int_{\mathbb{R}^2} C \cdot \mathbb{1}_{[0,2\pi]^2}(x, y) \,\mathrm{d}\lambda^2(x, y) \\ &= C\lambda^2([0, 2\pi]^2) = 4\pi^2 C < \infty \end{aligned}$$

The same holds for $\mathrm{Im}(F)$. Since the function F is absolutely integrable on $[0, 2\pi]^2$ we can use Fubini's theorem. This theorem also ensures, that

$\int_0^{2\pi} f(y)g(x-y)e^{-ikx}\,\mathrm{d}y$ is measurable wherever the integral exist. Moreover, we can say that the integral exists for all y, since the integrand is uniformly bounded.

$$\begin{aligned}\int_0^{2\pi}\int_0^{2\pi} F(x,y)\,\mathrm{d}y\,\mathrm{d}x &= \int_0^{2\pi}\int_0^{2\pi} F(x,y)\,\mathrm{d}x\,\mathrm{d}y \\ &= \int_0^{2\pi}\int_0^{2\pi} f(y)g(x-y)e^{-iky}e^{-ik(x-y)}\,\mathrm{d}x\,\mathrm{d}y \\ &= \int_0^{2\pi} f(y)e^{-iky}\int_0^{2\pi} g(x-y)e^{-ik(x-y)}\,\mathrm{d}x\,\mathrm{d}y\end{aligned}$$

Do the substitution $t = x - y$, $\frac{\partial t}{\partial x} = 1$ (this substitution is a diffeomorphism as a mapping $\Phi : [0; 2\pi] \to [-y, 2\pi - y]$, $x \mapsto x - y$ with $|\det(\mathrm{J}\,\Phi(x))| = 1$)

$$\begin{aligned}\int_0^{2\pi} f(y)e^{-iky}\int_0^{2\pi} g(x-y)e^{-ik(x-y)}\,\mathrm{d}x\,\mathrm{d}y &= \\ \int_0^{2\pi} f(y)e^{-iky}\int_{-y}^{2\pi-y} g(t)e^{-ikt}\,\mathrm{d}t\,\mathrm{d}y &= \\ \int_0^{2\pi} f(y)e^{-iky}\int_0^{2\pi} g(t)e^{-ikt}\,\mathrm{d}t\,\mathrm{d}y &= \\ 2\pi\int_0^{2\pi} f(y)e^{-iky}\hat{g}(k)\,\mathrm{d}y = 4\pi^2\hat{g}(k)\hat{f}(k)\end{aligned}$$

Thus:

$$2\pi\widehat{f*g}(k) = 4\pi^2\hat{g}(k)\hat{f}(k) \Rightarrow \frac{1}{2\pi}\widehat{f*g}(k) = \hat{g}(k)\hat{f}(k)$$

□

Theorem 4.2 (Discrete convolution theorem)**.** *Let $x, y \in \mathbb{R}^n$. Then the following holds:*

$$\frac{1}{\sqrt{N}}(F(x \circledast y)) = Fx \odot Fy$$

where $\circledast$ denotes the circular convolution, and $\odot$ is the component-wise multiplication

Proof. This is the direct analogy to the continuous case in Theorem 4.1 □

4.2 Ideal LPF

Using this knowledge we can derive an ideal low pass filter.

$$z_i = \begin{cases} 0, \ i \in (k; n-k) \\ 1, \ \text{otherwise} \end{cases}$$

This is an impulse reponse of the filter cutting all the frequencies above the k-th one. Recall that the $N-k$ element in DFT vector corresponds to the negative oscillation with the frequency k, due to cyclicity of $e^{\frac{2\pi ijx}{N}}$.

Using the formula for IDFT:

$$y_j = \sum_{l=0}^{N-1} F_{jl} z_k = \sum_{l=0}^{k} F_{jl} + \sum_{l=N-k}^{N-1} F_{jl} = \frac{1}{\sqrt{N}} \left(\sum_{l=0}^{k} \omega_N^{jl} + \sum_{l=N-k}^{N-1} \omega_N^{jl} \right)$$

Substitute $l = N - m - 1$ in the second sum:

$$\frac{1}{\sqrt{N}} \left(\sum_{l=0}^{k} \omega_N^{jl} + \sum_{l=N-k}^{N-1} \omega_N^{jl} \right) = \frac{1}{\sqrt{N}} \left(\sum_{l=0}^{k} \omega_N^{jl} + \sum_{m=0}^{k-1} \omega_N^{j(N-m-1)} \right) =$$

$$\frac{1}{\sqrt{N}} \left(\sum_{l=0}^{k} \omega_N^{jl} + \sum_{m=0}^{k-1} \omega_N^{j(N-m-1)} \right) = \frac{1}{\sqrt{N}} \left(\sum_{l=0}^{k} \omega_N^{jl} + \sum_{m=0}^{k-1} \omega_N^{jN} \omega_N^{-j} \omega_N^{-jm} \right) \underset{2.1.1}{=}$$

$$\frac{1}{\sqrt{N}} \left(\omega_N^{jk} + \sum_{l=0}^{k-1} \omega_N^{jl} + \omega_N^{-j} \sum_{m=0}^{k-1} \omega_N^{-jm} \right)$$

Let $j \neq 0$. Using the sum of geometric series:

$$\sum_{l=0}^{k-1} \omega_N^{jl} = \sum_{l=0}^{k-1} \left(\omega_N^{j}\right)^l = \frac{1 - \left(\omega_N^{j}\right)^k}{1 - \omega_N^{j}}$$

Similarly:

$$\omega_N^{-j} \sum_{m=0}^{k-1} \omega_N^{-jm} = \omega_N^{-j} \sum_{m=0}^{k-1} \left(\omega_N^{-j}\right)^m = \frac{\omega_N^{-j} - \omega_N^{-j} \left(\omega_N^{-j}\right)^k}{1 - \omega_N^{-j}} =$$

$$\frac{\omega_N^{-j} - \omega_N^{-j} \left(\omega_N^{-j}\right)^k}{1 - \omega_N^{-j}} \frac{\omega_N^{j}}{\omega_N^{j}} = \frac{1 - \left(\omega_N^{-j}\right)^k}{\omega_N^{j} - 1} = \frac{\left(\omega_N^{-j}\right)^k - 1}{1 - \omega_N^{j}}$$

By combining the terms together:

$$\sum_{l=0}^{k-1} \omega_N^{jl} + \omega_N^{-j} \sum_{l=0}^{k-1} \omega_N^{-jl} = \frac{1 - \left(\omega_N^j\right)^k}{1 - \omega_N^j} + \frac{\left(\omega_N^{-j}\right)^k - 1}{1 - \omega_N^j} = \frac{\omega^{-jk} - \omega^{jk}}{1 - \omega_N^j}$$

This yields the following expression for y_j, when $j \neq 0$:

$$y_j = \frac{1}{\sqrt{N}} \left(\omega_N^{jk} + \frac{\omega_N^{-jk} - \omega_N^{jk}}{1 - \omega_N^j} \right)$$

If $j = 0$, then all the summands are equal to one, so get just $(2k+1)/\sqrt{N}$. By the Theorem 4.2:

$$\frac{1}{\sqrt{N}}(F(x \circledast y)) = Fx \odot Fy$$

which allows us to implement the low pass filter using its impulse response. That is the point, where theory meets the real world example and makes use of the abstract theorem. In general this concept can be generalaized to any signal, which can be used, for instance for reverb implementation if we have an impulse response of a room (a recording of echo induced by a short impact).

Now we take a closer look on the expression we had derived above:

$$e^{-ix} - e^{ix} = 2\,\mathrm{Im}(e^{-ix}) = 2i\sin(-x) = -2i\sin(x)$$

$$1 - e^{ix} = e^{\frac{ix}{2}} \left(e^{-\frac{ix}{2}} - e^{\frac{ix}{2}} \right) = -2i \cdot e^{\frac{ix}{2}} \sin\left(\frac{x}{2}\right)$$

Therefore:

$$1 - \omega_N^j = 1 - e^{2\pi i \frac{j}{N}} = -2i \cdot e^{\frac{\pi i \cdot j}{N}} \sin\left(\frac{\pi j}{N}\right)$$

$$\omega_N^{-jk} - \omega_N^{jk} = e^{-2\pi i \frac{kj}{N}} - e^{2\pi i \frac{kj}{N}} = -2i \sin\left(\frac{2\pi kj}{N}\right)$$

$$\frac{\omega_N^{-jk} - \omega_N^{jk}}{1 - \omega_N^j} = \frac{-2i \sin\left(\frac{2\pi kj}{N}\right)}{-2i \cdot e^{\frac{\pi i \cdot j}{N}} \sin\left(\frac{\pi j}{N}\right)} = \frac{e^{-\frac{\pi i \cdot j}{N}} \sin\left(\frac{2\pi kj}{N}\right)}{\sin\left(\frac{\pi j}{N}\right)}$$

With $x_j = 2\pi j/N$ we have:

$$\frac{e^{-\frac{\pi i \cdot j}{N}} \sin\left(\frac{2\pi k j}{N}\right)}{\sin\left(\frac{\pi j}{N}\right)} = \frac{e^{-ix_j/2} \sin\left(kx_j\right)}{\sin\left(x_j/2\right)} = \frac{\cos(x_j/2)\sin(kx_j)}{\sin(x_j/2)} - i\sin\left(kx_j\right)$$

Add the term ω_N^{jk} we had previously:

$$\begin{aligned} \omega_N^{jk} + \frac{\cos(x_j/2)\sin(kx_j)}{\sin(x_j/2)} - i\sin(kx_j) = \\ \cos(kx_j) + i\sin(kx_j) + \frac{\cos(x_j/2)\sin(kx_j)}{\sin(x_j/2)} - i\sin(kx_j) = \\ \cos(kx_j) + \frac{\cos(x_j/2)\sin(kx_j)}{\sin(x_j/2)} = \\ \frac{\cos(kx_j)\sin(x_j/2) + \cos(x_j/2)\sin(kx_j)}{\sin(x_j/2)} \end{aligned} \tag{8}$$

Using the equality $\sin(x+y) = \sin(x)\cos(y) + \sin(y)\cos(x)$:

$$(8) = \frac{\sin((k+1/2)x_j)}{\sin(x_j/2)}$$

$$y_j = \begin{cases} \frac{1}{\sqrt{N}} \frac{\sin((k+1/2)x_j)}{\sin(x_j/2)}, \ j \in \{1, \dots, N-1\} \\ (2k+1)/\sqrt{N}, \ j = 0 \end{cases}$$

Miraculously we got a continuous Dirichlet kernel, discretized at points x_j. One can show, that the impulse response of Dirichlet kernel in continuous space acts in the exact same way, namely cuts the frequencies above the k-th one. As we had seen in equation (2), $z_k = \sum_{n\in\mathbb{Z}} \hat{f}(k+nN)$, so given that $\widehat{D_n}(k) = \begin{cases} 1, k \leq n \\ 0, \text{ otherwise} \end{cases}$, this result was not completely unexpected.

5 Infinite impulse response filters

We kick off with the example

Example 5.1. Consider the sequence $x[n] \in \mathbb{R}$, which represents the signal evolving in time. Let:

$$y[0] = x[0], \quad y[n] = x[n] + \alpha y[n-1] \; \forall n > 0$$

The structure of the formula suggests, that we should use an induction to get a closed formula for y:

Base. $y[0] = x[0] = \alpha^0 x[0]$

Step. Assume that it holds for some n, that

$$y[n] = \sum_{j=0}^{n} \alpha^{n-j} x[j] \tag{9}$$

Then:

$$\begin{aligned} y[n+1] = x[n+1] + \alpha y[n] &= \alpha^0 x[n+1] + \alpha \sum_{j=0}^{n} \alpha^{n-j} x[j] \\ &= \alpha^0 x[n+1] + \sum_{j=0}^{n} \alpha^{n+1-j} x[j] = \sum_{j=0}^{n+1} \alpha^{n+1-j} x[j] \end{aligned}$$

Note that the formula (9) can be written as a following convolution:

$$y[n] = \sum_{j=0}^{n} \alpha^{n-j} x[j] = \sum_{j=0}^{n} a_n[n-j] x[j] = x * a_n$$

Thus:

$$a_n = \left(\alpha^0, \alpha^1, \ldots, \alpha^n\right)^T$$

As in the case of Ideal low pass filter in Section 4.2 we got an impulse response of the filter introduced above. The only difference is that this filter has an impulse response with the size proportional to the sample size. That said the IR size grows infinitely in time unlike the IR of an ideal LPF, which has the same size for each frame.

We make an assumption on x, that its series is absolutely convergent. That is

$$\sum_{n=0}^{\infty} |x[n]| < \infty$$

Then for $n > m$ we have the following estimate for $0 < \alpha < 1$:

$$\begin{aligned}
|y[n] - y[m]| &= \left| \sum_{j=0}^{n} \alpha^{n-j} x[j] - \sum_{j=0}^{m} \alpha^{m-j} x[j] \right| \\
&= \left| \sum_{j=0}^{m} (\alpha^{n-j} - \alpha^{m-j}) x[j] + \sum_{j=m+1}^{n} \alpha^{n-j} x[j] \right| \\
&\leq \left| \sum_{j=0}^{m} (\alpha^{n-j} - \alpha^{m-j}) x[j] \right| + \left| \sum_{j=m+1}^{n} \alpha^{n-j} x[j] \right|
\end{aligned}$$

Since the sum $\sum_{n=0}^{\infty} |x[n]|$ converges we have that x_n is the zero sequence, hence for any $\varepsilon > 0$ exists n_0, s.t. $|x_n| < \varepsilon \; \forall n \geq n_0$:

$$\left| \sum_{j=0}^{m} (\alpha^{n-j} - \alpha^{m-j}) x[j] \right| \leq \sum_{j=0}^{m} |\alpha^{n-j} - \alpha^{m-j}| \cdot |x[j]| =$$

$$\sum_{j=0}^{n_0} |\alpha^{n-j} - \alpha^{m-j}| \cdot |x[j]| + \sum_{j=n_0}^{m} |\alpha^{n-j} - \alpha^{m-j}| \cdot |x[j]| \leq$$

$$\sum_{j=0}^{n_0} |\alpha^{n-j} - \alpha^{m-j}| \cdot |x[j]| + \varepsilon \sum_{j=n_0}^{m} |\alpha^{n-j} - \alpha^{m-j}|$$

With the assumption that $n > m$ we have $n - j > m - j$, so by the monotonicty of α^x we have that $\alpha^{n-j} < \alpha^{m-j} \Rightarrow |\alpha^{n-j} - \alpha^{m-j}| = \alpha^{m-j} - \alpha^{n-j} \leq \alpha^{m-j}$. Moreover $\sum_{j=n_0}^{m} \alpha^{m-j} \leq \sum_{j=0}^{\infty} \alpha^j = C \leq \infty$ since $m - j$ stays positive on the entire range of summation. For the same exact reason $|\alpha^{n-j} - \alpha^{m-j}| = \alpha^{m-j} - \alpha^{n-j} \leq \alpha^{m-n_0}$ if $j \leq n_0$. By combining those two estimates together:

$$\sum_{j=0}^{n_0} |\alpha^{n-j} - \alpha^{m-j}| \cdot |x[j]| + \varepsilon \sum_{j=n_0}^{m} |\alpha^{n-j} - \alpha^{m-j}| \leq$$

$$\alpha^{m-n_0} \sum_{j=0}^{n_0} |x[j]| + C\varepsilon = \alpha^{m-n_0} M + C\varepsilon \to 0$$

as we can choose such m, that $\alpha^{m-n_0}M < \varepsilon$, so the whole expression is bounded by $(C+1)\varepsilon$

For the second term we have that:

$$\left|\sum_{j=m+1}^{n} \alpha^{n-j}x[j]\right| \leq \sum_{j=m+1}^{n} |\alpha^{n-j}||x[j]| \leq \sum_{j=m+1}^{n} |x[j]|$$

which can be made arbitrarily small as the tail sum of a convergent sequence.

This proves for that sufficiently large m we can get $|y[n]-y[m]|$ arbitrarily small, so $y[n]$ is a Cauchy sequence and hence converges.

FIX THIS ENTIRELY

Observe that the way we proved the filter output to be absolutely summable became really tedious despite the simplicty of its definition. That motivates us develop the set of tools to analyze such filters relatively easy.

5.1 Stability

Let us consider the general form of the filters we would like to analyze:

$$y[n] = \sum_{i=0}^{N} b_i x[n-i] + \sum_{i=1}^{M} a_i y[n-i] \tag{10}$$

Lemma 5.1 (Z-transform properties)**.** *Let $x \overset{\mathcal{Z}}{\leftrightarrow} X(z)$ and $y \overset{\mathcal{Z}}{\leftrightarrow} Y(z)$. The following holds:*

5.1.1 $x[n-k] \overset{\mathcal{Z}}{\leftrightarrow} z^{-k}X(z)$

5.1.2 $\alpha x[n] + \beta y[n] \overset{\mathcal{Z}}{\leftrightarrow} \alpha X(z) + \beta Y(z)$

Proof.

1.
$$\begin{aligned}\sum_{n=-\infty}^{\infty} x[n-k]z^{-n} &= \sum_{m=-\infty}^{\infty} x[m]z^{-(m+k)} \\ &= z^{-k} \sum_{m=-\infty}^{\infty} x[m]z^{-m} \\ &= z^{-k}X(z)\end{aligned}$$

2. We assume that both $X(z)$ and $Y(z)$ converge for the given value of z

$$\begin{aligned}\sum_{n=-\infty}^{\infty} (\alpha x[n] + \beta y[n])z^{-n} &= \sum_{n=-\infty}^{\infty} \alpha x[n]z^{-n} + \beta y[n]z^{-n} \\ &= \alpha \sum_{n=-\infty}^{\infty} x[n]z^{-n} + \beta \sum_{n=-\infty}^{\infty} y[n]z^{-n} \\ &= \alpha X(z) + \beta Y(z)\end{aligned}$$

□

Taking the Z-transform of (10):

$$y[n] \overset{\mathcal{Z}}{\leftrightarrow} \sum_{i=0}^{N} b_i z^{-i} X(z) + \sum_{i=1}^{M} a_i z^{-i} Y(z)$$

That is:

$$Y(z) = \sum_{i=0}^{N} b_i z^{-i} X(z) + \sum_{i=1}^{M} a_i z^{-i} Y(z)$$

$$Y(z) - \sum_{i=1}^{M} a_i z^{-i} Y(z) = \sum_{i=0}^{N} b_i z^{-i} X(z)$$

$$\frac{Y(z)}{X(z)} = \frac{\sum_{i=0}^{N} b_i z^{-i}}{1 - \sum_{i=1}^{M} a_i z^{-i}}$$

This is called a transfer function and it is crucial for analyzing the filter's behaviour. Recall the definition of DFT:

$$z_k = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kj} y_k = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \left(e^{\frac{2\pi i j}{N}} \right)^{-k} y_k$$

Now we denote $z = e^{\frac{2\pi i j}{N}}$ and drop the normalization factor in front. The expression becomes:

$$\sum_{k=0}^{N-1} z^{-k} y_k$$

is exactly the Z-transform of the signal y (embedded into the infinite sequence). In general as N gets large we may assume that for any $\omega \in [0, 2\pi]$ exists $j \in \mathbb{N}$ such that $\frac{2\pi j}{N} \approx \omega$, which allows us to compute the presence of any frequency in the range $[0, 2\pi]$ by evaluating the signal Z-transform at a point $z = e^{i\omega}$. Moreover the Theorem 4.2 suggests that the function $H(z)$ represents an impulse response as it essentially does deconvolution in the frequency (z) domain.

Lemma 5.2 (Partial fraction expansion)**.** *Let $P(x) \in \mathbb{C}_M[x]$, $a_i \in \mathbb{C}$ for $i \in \{1, \ldots, N\}$ and $M < N$*

$$Q(x) = \frac{P(x)}{\prod_{k=1}^{N} (1 - a_k x)}$$

with $a_i \neq a_j \ \forall i \neq j$ and $a_i \neq 0$.

Then the function $Q(x)$ can be expressed in the following form:

$$Q(x) = \sum_{i=1}^{N} \frac{A_i}{(1 - a_i x)}$$

and moreover this representation is unique.

Proof. Let $Q_i(x) = \prod_{k \neq i}^{N}(1 - a_k x)$. Consider their linear combination:

$$\sum_{i=1}^{N} \lambda_i Q_i(x) = 0$$

Substitute a_j^{-1}, which is well defined since $a_j \neq 0 \ \forall j \in \mathbb{N}$. We observe that $Q_i(a_j^{-1}) = 0$ for $i \neq j$ since Q_i contains the term $(1 - a_j x)$. Also $a_i a_j^{-1} \neq 1 \ \forall i \neq j$ since $a_i \neq a_j$, so $Q_j(a_j^{-1}) \neq 0$. Then:

$$\sum_{i=1}^{N} \lambda_i Q_i(a_j^{-1}) = \lambda_j Q_j(a_j^{-1}) = 0$$

Since the field $(\mathbb{C}, +, \cdot)$ is zero divisor free we must have $\lambda_j = 0$. This holds for all $j \in \{1, \ldots, N\}$.

Thus Q_i's form the basis of $\mathbb{C}_{N-1}[x]$ as $\dim \mathbb{C}_{N-1}[x] = N$, hence in particular we can decompose every polynomial in $\mathbb{C}_M[x]$ using the linear combination of Q_i:

$$\frac{P(x)}{\prod_{k=1}^{N}(1 - a_k x)} = \frac{\sum_{i=1}^{N} A_i Q_i}{\prod_{k=1}^{N}(1 - a_k x)} = \sum_{i=1}^{N} \frac{A_i Q_i}{\prod_{k=1}^{N}(1 - a_k x)} = \sum_{i=1}^{N} \frac{A_i}{1 - a_i x}$$

Thus we proved the existence of partial fraction decomposition whenever the numerator degree is less than the degree of denominator and denominator does not contain zero roots. Uniqueness follows from the basis properties. □

Lemma 5.3 (Long division)**.** *Let $P \in C_N[x]$ and $Q \in C_M[x]$ with $N \geq M > 0$ and $\deg P = N$, $\deg Q = M$. Then there exists a polynomial $q \in C_{N-M}[x]$ and $r \in C_{M-1}[x]$, s.t. $P = q \cdot Q + r$*

Proof. Let $P(x) = a_N x^N + \cdots + a_1 x + a_0$ and $Q(x) = b_M x^M + \cdots + b_1 x + b_0$ with $a_N \neq 0$ and $b_M \neq 0$ to ensure that polynomials have the proper degree. We prove this statement by induction:

Base: Let $N = M$. Set $q(x) = a_N / b_M$ and $r(x) = P(x) - q(x) \cdot Q(x)$.T his fraction is well defined since $b_M \neq 0$. $P = q \cdot Q + r$ follows by

construction, so we need to show that $\deg r(x) \leq M$:

$$r(x) = a_N x^N + \cdots + a_1 x + a_0 - \frac{a_N}{b_M}(b_M x^M + \cdots + b_1 x + b_0) =$$
$$a_N x^N + \cdots + a_1 x + a_0 - a_N x^M - \frac{a_N b_{M-1}}{b_M} x^{M-1} - \cdots - \frac{a_N b_0}{b_M} \underset{N=M}{=}$$
$$a_N x^N + \cdots + a_1 x + a_0 - a_N x^N - \frac{a_N b_{N-1}}{b_N} x^{N-1} - \cdots - \frac{a_N b_0}{b_N} =$$
$$a_{M-1} x^{M-1} + \cdots + a_1 x + a_0 - \frac{a_M b_{M-1}}{b_N} x^{M-1} - \cdots - \frac{a_M b_0}{b_M}$$

That is $\deg r \leq M - 1$

Step: Suppose the statement holds for some $N > M$. Then:

$$P(x) = a_{N+1} x^{N+1} + \cdots + a_1 x + a_0 = x(a_{N+1} x^N + \cdots + a_2 x + a_1) + a_0$$
$$\underset{(a)}{=} x(q(x) \cdot Q(x) + r(x)) + a_0 = (xq(x)) \cdot Q(x) + (xr(x) + a_0)$$

where (a) follows from the induction hypothesis, therefore $\deg q \leq N - M$ and $\deg r \leq M - 1$.

Note that $\deg xq \leq N + 1 - M$ and $\deg xr \leq M$, $\deg a_0 = 0$. If the degree of $xr(x) + a_0$ is exactly M we may use the base case to find the decomposition $xr(x) = \tilde{q}(x)Q(x) + \tilde{r}(x)$ where $\deg \tilde{q} = 0$ and $\deg \tilde{r} \leq M - 1$. Then $q^*(x) = xq(x) + \tilde{q}(x)$, $\deg q^* \leq N + 1 - M$ and $P(x) = q^*(x)Q(x) + \tilde{r}(x)$

□

Theorem 5.1. *Let $X(z)$ be a z-tranform of a causal system $x[n]$ (i.e. $x[n] = 0$, $\forall n < 0$). Moreover assume that X is the rational function with distinct non-zero poles. Then $x[n]$ is absolutely summable if the pole with the maximum magnitude lies inside the unit circle.*

Proof.

$$X(z) = \frac{P(z^{-1})}{Q(z^{-1})}$$

If $\deg P = N > M = \deg Q$, then the Lemma 5.3 ensures that exist $q(z^-1), r(z^-1)$ with $\deg q \leq N - M$ and $\deg r \leq M - 1$, s.t. $P(z^{-1}) = q(z^{-1})Q(z^{-1}) + r(z^{-1})$.

$$X(z) = \frac{q(z^{-1})Q(z^{-1}) + r(z^{-1})}{Q(z^{-1})} = q(z^{-1}) + \frac{r(z^{-1})}{Q(z^{-1})}$$

Let $d_1, \ldots, d_M$ be the poles of X (distinct and non-zero by assumption), so $Q(z^{-1}) = \prod_{i=1}^{M}(1 - d_i z^{-1})$

Using the Lemma 5.2 we have that exist $A_1, \ldots, A_M$, such that:

$$\frac{r(z^{-1})}{Q(z^{-1})} = \sum_{i=1}^{M} \frac{A_i}{1 - d_i z^{-1}}$$

$$X(z) = q(z^{-1}) + \sum_{i=1}^{M} \frac{A_i}{1 - d_i z^{-1}} = \sum_{i=0}^{N-M} c_i z^{-i} + \sum_{i=1}^{M} \frac{A_i}{1 - d_i z^{-1}}$$

The assumption about causality allows us to uniquelly determine the inverse z-transform by inspecting the table.

$$c_i z^{-i} \overset{\mathcal{Z}^{-1}}{\leftrightarrow} c_i \delta[n - i]$$

$$\frac{A_i}{1 - d_i z^{-1}} \overset{\mathcal{Z}^{-1}}{\leftrightarrow} A_i d_i^n u[n]$$

Using the Lemma 5.1.2:

$$\sum_{i=0}^{N-M} c_i z^{-i} + \sum_{i=1}^{M} \frac{A_i}{1 - d_i z^{-1}} \overset{\mathcal{Z}^{-1}}{\leftrightarrow} \sum_{i=0}^{N-M} c_i \delta[n - i] + \sum_{i=1}^{M} A_i d_i^n u[n]$$

$$\sum_{n=0}^{\infty} |x[n]| = \sum_{n=-\infty}^{\infty} \left| \sum_{i=0}^{N-M} c_i \delta[n - i] + \sum_{i=1}^{M} A_i d_i^n u[n] \right|$$

Note that $\delta[n - i] = 1$ if and only if $n = i$ and zero elsewhere.

$$\begin{aligned} \sum_{n=-\infty}^{\infty} |x[n]| &\leq \sum_{n=-\infty}^{\infty} \sum_{i=0}^{N-M} |c_i \delta[n - i]| + \sum_{i=1}^{M} |A_i d_i^n u[n]| \\ &= \sum_{i=0}^{N-M} |c_i| + \sum_{n=-\infty}^{\infty} \sum_{i=1}^{M} |A_i d_i^n u[n]| \end{aligned}$$

We require each sum $\sum_{n=-\infty}^{\infty} |A_i d_i^n u[n]|$ for $\{1, \ldots, M\}$ to converge, as this would imply that $\sum_{n=-\infty}^{\infty} \sum_{i=1}^{M} |A_i d_i^n u[n]|$ converges as well by limit additivity.

$$\sum_{n=-\infty}^{\infty} |A_i d_i^n u[n]| = |A_i| \sum_{n=0}^{\infty} |d_i^n| < \infty$$

This is the sum of geometric series, hence converges for d_i with $|d_i| < 1$. We need this to hold for all $i \in \{1, \ldots, M\}$, so $\max_i |d_i| < 1$ $\square$

We say that z-transform converges if the series

$$\sum_{n=-\infty}^{\infty} |x[n]z^{-n}| = \sum_{n=-\infty}^{\infty} |x[n]|r^{-n}$$

converges (reference openheim 3.6 for completeness). It can be easily seen that if the z-transform converges for some point located at the distance r to the origin, then it converges for any point at the circle with radius r centered at the origin. This immediately yields the following equivalence:

$$X(e^{i\omega}) \text{ converges } \forall \omega \in [0, 2\pi] \Leftrightarrow x[n] \text{ absolutely summable}$$

Return to the Example 5.1:

$$y[n] = x_n + \alpha y[n-1]$$

Its transfer function is:

$$H(z) = \frac{1}{1 - \alpha z^{-1}}$$

which has the only pole at $z = \alpha$.

proper justification of y absolute summability

5.2 Notch filter

We want to construct the transfer function with the zeros corresponding to the frequency ω. Its numerator then is the function $(z^{-1} - e^{i\omega})(z^{-1} - e^{-i\omega})$ since we need to cancel both positive and negative frequency oscillations. If we define the transfer function as is the frequency response looks as follows:

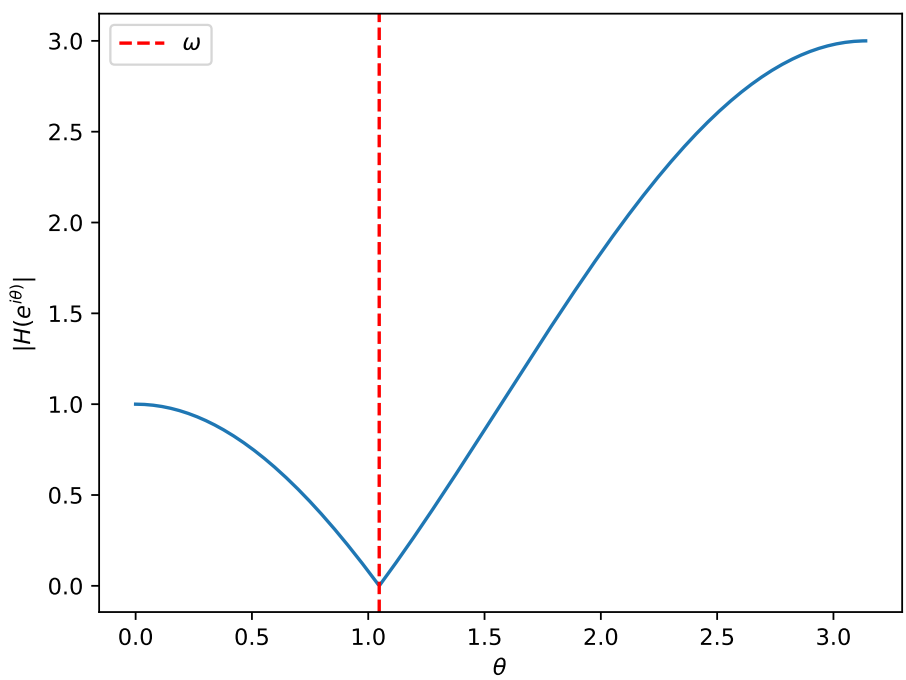

Figure 3: the frequency response of $(z^{-1} - e^{i\omega})(z^{-1} - e^{-i\omega})$

As we can see from the Figure 3 this filter produces a significant uneveness in the spectrum. Moreover we cannot really control the steepness of the constructed filter. To mitigate this we divide the transfer function we got previously by $(rz^{-1} - e^{i\omega})(rz^{-1} - e^{-i\omega})$. This helps to flatten the filter since $rz^{-1} - e^{i\omega} \approx (z^{-1} - e^{i\omega})$ for r reasonably close to 1. Note that dividing the transfer function by this expression adds two poles at $d_1 = re^{-i\omega}$ and $d_2 = re^{i\omega}$, so for the filter to be stable (converge for the unit circle) the radius r must be less that 1 by the Theorem 5.1. The bigger the value of r,

the closer the pole is to the zeros, so it makes the filter steeper as it "lifts" the surface around it.

Define:

$$H(z) = \frac{(z^{-1} - e^{i\omega})(z^{-1} - e^{-i\omega})}{(rz^{-1} - e^{i\omega})(rz^{-1} - e^{-i\omega})} = \frac{z^{-2} - (e^{-i\omega} + e^{i\omega})z^{-1} + 1}{r^2 z^{-2} - r(e^{-i\omega} + e^{i\omega})z^{-1} + 1}$$
$$= \frac{z^{-2} - 2\cos(\omega)z^{-1} + 1}{r^2 z^{-2} - 2r\cos(\omega)z^{-1} + 1}$$

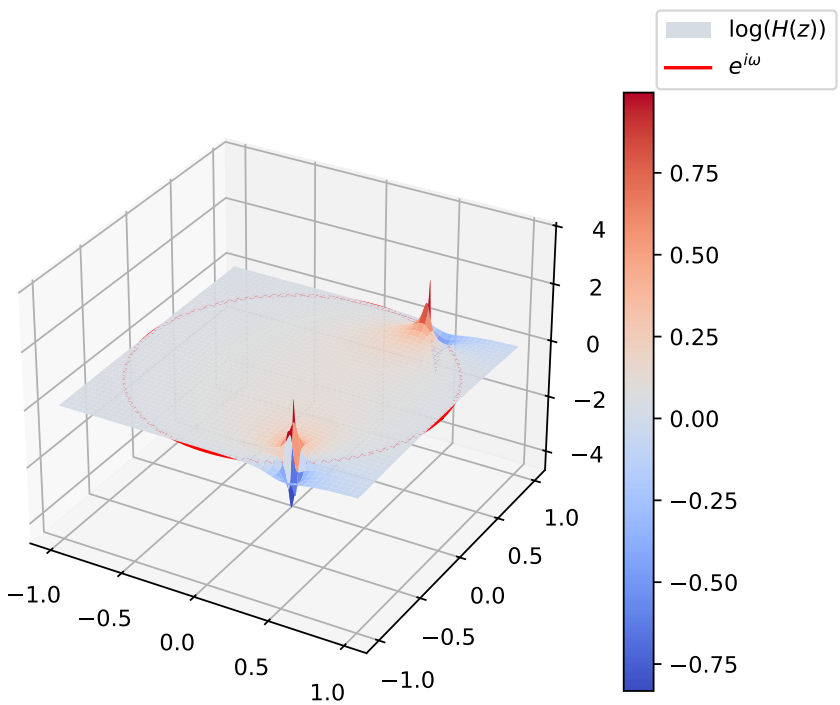

Figure 4: the filter topology

Note that there are two spike slightly inside of the red circle and two dips. Those are the poles and the zeros of H.

And its frequency response looks as follows:

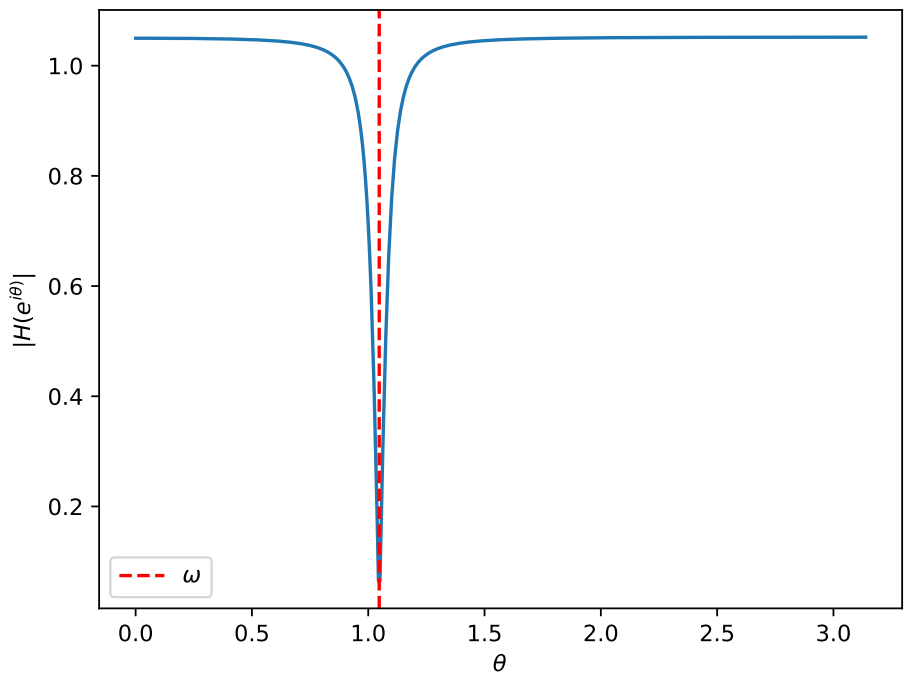

Figure 5: the filter frequency response

Note that for the transfer function $H(z)$ we have:

$$\overline{H(z)} = \overline{\frac{z^{-2} - 2\cos(\omega)z^{-1} + 1}{r^2 z^{-2} - 2r\cos(\omega)z^{-1} + 1}} = \frac{\overline{z^{-2}} - 2\cos(\omega)\overline{z^{-1}} + 1}{r^2\overline{z^{-2}} - 2r\cos(\omega)\overline{z^{-1}} + 1} = H(\overline{z})$$

That means for the frequency $e^{-i\theta}$ the frequency response has the same magnitude as for $e^{i\theta}$, but inverted phase. By the Remark 2.1.2, we have that this directly relates to the corresponding oscillation phase. This allows us the visualize also the phase response of the constructed filter:

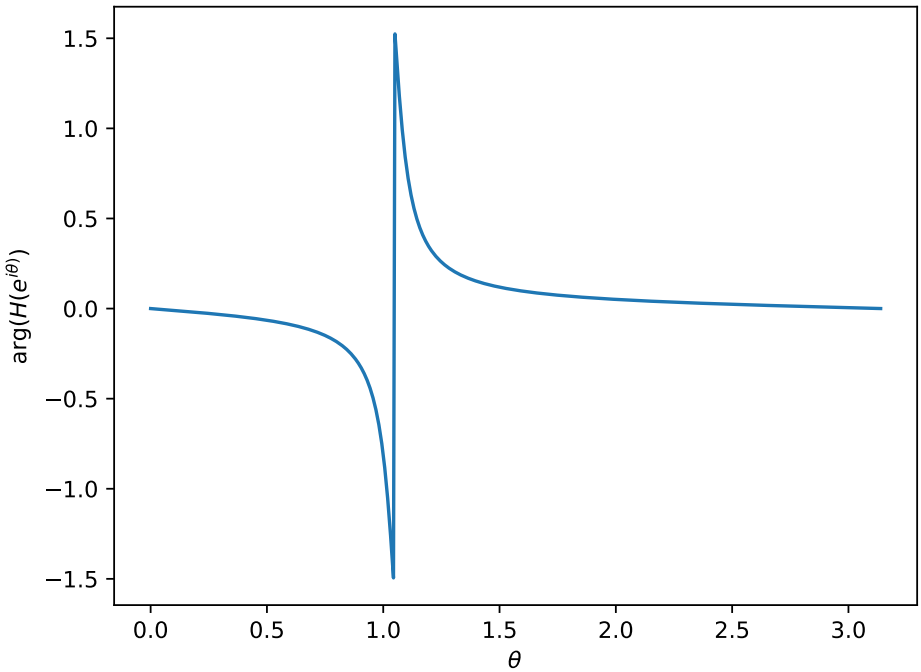

Figure 6: the filter phase response

6 Appendix

6.1 Notch filter impulse response 🗑

We perform the long division of the numerator and denominator:

$$z^{-2} - (e^{-i\omega} + e^{i\omega})z^{-1} + 1 = \frac{1}{r^2}(r^2 z^{-2} - r(e^{-i\omega} + e^{i\omega})z^{-1} + 1) + \frac{(e^{-i\omega} + e^{i\omega})z^{-1}}{r} - \frac{1}{r^2} - (e^{-i\omega} + e^{i\omega})z^{-1} + 1$$

so the transfer function becomes:

$$\begin{aligned} H(z) &= \frac{1}{r^2} + \frac{\frac{(e^{-i\omega}+e^{i\omega})z^{-1}}{r} - \frac{1}{r^2} - (e^{-i\omega} + e^{i\omega})z^{-1} + 1}{r^2 z^{-2} - r(e^{-i\omega} + e^{i\omega})z^{-1} + 1} \\ &= \frac{1}{r^2} + \frac{(e^{-i\omega} + e^{i\omega})\left(\frac{1}{r} - 1\right)z^{-1} - \frac{1}{r^2} + 1}{(rz^{-1} - e^{i\omega})(rz^{-1} - e^{-i\omega})} \end{aligned}$$

Using the partial fraction expansion discussed in Lemma 5.2:

$$\begin{aligned} H(z) &= \frac{1}{r^2} + \frac{A}{rz^{-1} - e^{i\omega}} + \frac{B}{rz^{-1} - e^{-i\omega}} \\ &= \frac{1}{r^2} + \frac{A(rz^{-1} - e^{-i\omega}) + B(rz^{-1} - e^{i\omega})}{(rz^{-1} - e^{i\omega})(rz^{-1} - e^{-i\omega})} \end{aligned}$$

We get the following equation:

$$A(rz^{-1} - e^{-i\omega}) + B(rz^{-1} - e^{i\omega}) = (e^{-i\omega} + e^{i\omega})\left(\frac{1}{r} - 1\right)z^{-1} - \frac{1}{r^2} + 1$$

$$r(A + B)z^{-1} - (Ae^{-i\omega} + Be^{i\omega}) = (e^{-i\omega} + e^{i\omega})\left(\frac{1}{r} - 1\right)z^{-1} - \frac{1}{r^2} + 1$$

Which result in the system of equation:

$$\begin{cases} r(A + B) = (e^{-i\omega} + e^{i\omega})\left(\frac{1}{r} - 1\right) \\ Ae^{-i\omega} + Be^{i\omega} = 1 - \frac{1}{r^2} \end{cases}$$

$$B = (e^{-i\omega} + e^{i\omega})\left(\frac{1}{r^2} - \frac{1}{r}\right) - A$$

$$
\begin{aligned}
Ae^{-i\omega} + Be^{i\omega} &= Ae^{-i\omega} + \left((e^{-i\omega} + e^{i\omega}) \left(\frac{1}{r^2} - \frac{1}{r} \right) - A \right) e^{i\omega} \\
&= A(e^{-i\omega} - e^{i\omega}) + (1 + e^{2i\omega}) \left(\frac{1}{r^2} - \frac{1}{r} \right)
\end{aligned}
$$

$$
A(e^{-i\omega} - e^{i\omega}) + (1 + e^{2i\omega}) \left(\frac{1}{r^2} - \frac{1}{r} \right) = 1 - \frac{1}{r^2}
$$

$$
A(e^{-i\omega} - e^{i\omega}) = 1 - \frac{1}{r^2} - (1 + e^{2i\omega}) \left(\frac{1}{r^2} - \frac{1}{r} \right)
$$

$$
A = \frac{1 - \frac{1}{r^2} - (1 + e^{2i\omega}) \left(\frac{1}{r^2} - \frac{1}{r} \right)}{e^{-i\omega} - e^{i\omega}} = \frac{1 - \frac{1}{r^2} - (1 + e^{2i\omega}) \left(\frac{1}{r^2} - \frac{1}{r} \right)}{e^{-i\omega} - e^{i\omega}}
$$

$$
\begin{aligned}
B &= (e^{-i\omega} + e^{i\omega}) \left(\frac{1}{r^2} - \frac{1}{r} \right) - \frac{1 - \frac{1}{r^2} - (1 + e^{2i\omega}) \left(\frac{1}{r^2} - \frac{1}{r} \right)}{e^{-i\omega} - e^{i\omega}} \\
&= \frac{(e^{-2i\omega} - e^{2i\omega}) \left(\frac{1}{r^2} - \frac{1}{r} \right)}{e^{-i\omega} - e^{i\omega}} - \frac{1 - \frac{1}{r^2} - (1 + e^{2i\omega}) \left(\frac{1}{r^2} - \frac{1}{r} \right)}{e^{-i\omega} - e^{i\omega}} \\
&= \frac{1 - \frac{1}{r^2} + (1 + e^{-2i\omega}) \left(\frac{1}{r^2} - \frac{1}{r} \right)}{e^{-i\omega} - e^{i\omega}}
\end{aligned}
$$

$$
\boxed{A = \frac{1 - \frac{1}{r^2} - (1 + e^{2i\omega}) \left(\frac{1}{r^2} - \frac{1}{r} \right)}{e^{-i\omega} - e^{i\omega}} \quad B = \frac{1 - \frac{1}{r^2} + (1 + e^{-2i\omega}) \left(\frac{1}{r^2} - \frac{1}{r} \right)}{e^{-i\omega} - e^{i\omega}}}
$$

Now we are able to compute the inverse transform using the fact that

$$
a^n u[n] \overset{\mathcal{Z}}{\leftrightarrow} \frac{1}{1 - az^{-1}}
$$

$$
\delta[n] \overset{\mathcal{Z}}{\leftrightarrow} 1
$$

$$
\begin{aligned}
H(Z) &= \frac{1}{r^2} + \frac{Ae^{-i\omega}}{re^{-i\omega}z^{-1} - 1} + \frac{Be^{i\omega}}{re^{i\omega}z^{-1} - 1} \\
&\overset{\mathcal{Z}^{-1}}{\leftrightarrow} \frac{1}{r^2}\delta[n] - Ar^n e^{-i\omega(n+1)} u[n] - Br^n e^{i\omega(n+1)} u[n]
\end{aligned}
$$

References

[1] Elias M. Stein and Rami Shakarchi (2003) *Fourier Analysis: An Introduction*